



Parking Monitoring System: LoRa Vision

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Motivation

- Edge computer vision often requires expensive hardware, or a reliable internet connection to offload image processing to a central server.
- LoRa technology offers long-range, low-power, wireless connectivity, but suffers from low bandwidth, which has limited its implementation for some data-intensive applications.
- **LoRaVSN** offers a framework for transferring image data through LoRa networks in near real-time, designed to extend computer vision capabilities to environments where high-bandwidth Internet access is limited, unreliable, or unavailable.





Specific Objectives

Transmission Framework

- Enable low-cost edge devices to be part of Computer Vision data pipelines using LoRa communication, and reducing traditional deployment costs by over 50%.

Data Reduction Pipeline

- Reduce image size by over 70% by using grayscale conversion and masks over regions-of-interests (ROIs) before transmission.

LoRa Communication

- Packetize image data into LoRa-compatible messages for transmission while maintaining successful image packet receptions over 80%.

Gateway Processing

- Receive and prepare image data for downstream computer vision tasks with processing times below 5 minutes per image.

PARKS Integration

- Validate LoRaVSN as the communication layer for the PARKS Parking Monitoring System.



Solution Approach



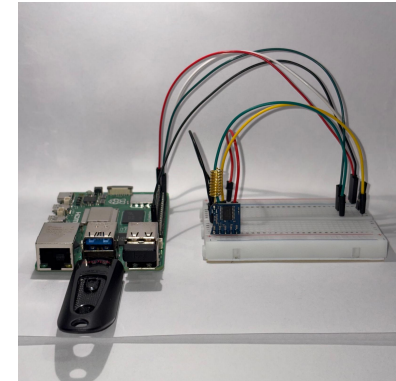
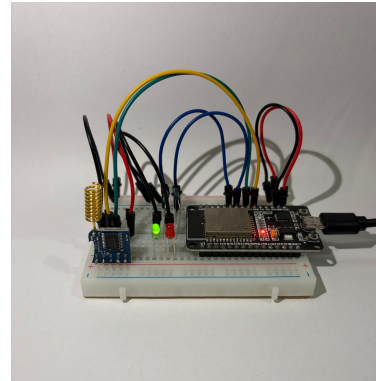
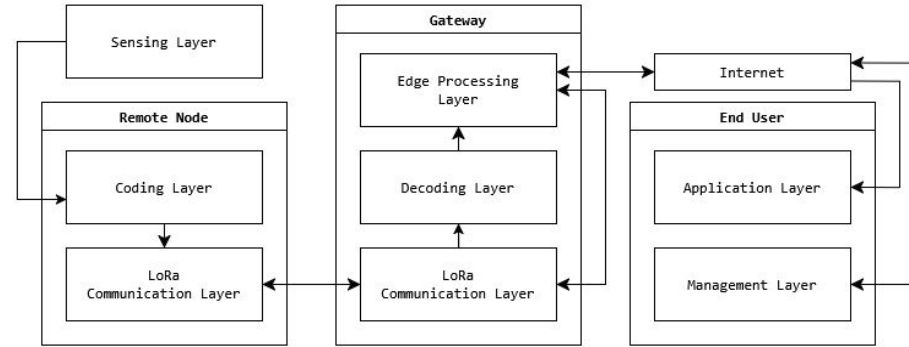
System Architecture

Remote Node

- Generic ESP32-WROOM-32 with 520KB onboard memory.
- REYAX RYLR998 LoRa transceiver.
- **Main Tasks:** ROI masking, JPEG compression, and packetized image transmission.

Gateway

- Raspberry Pi 5 with 8GB onboard memory.
- REYAX RYLR998 LoRa transceiver.
- Provides a Web HTTP API for integration.
- **Main Tasks:** Packet reassembly, image reconstruction, and data preparation.





Data Reduction: Grayscale & ROI Extraction

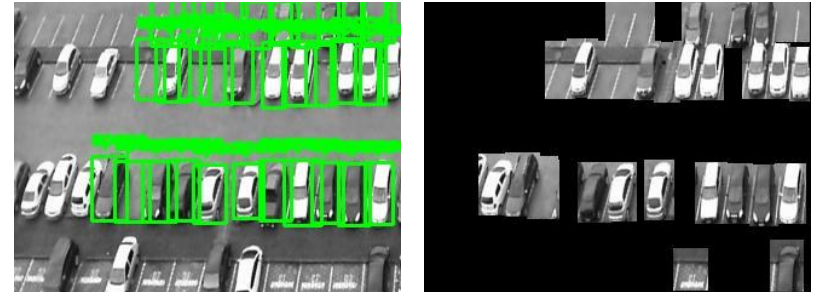
Step 1: Grayscale Conversion

- Sensing Layer captures RGB images, and provides grayscale bytes at 320x240 resolution.
- RGB images use 3 bytes per pixel and are converted to grayscale using 1 byte per pixel.
- Size reduction: 230,400 bytes to 76,800 bytes (~66% reduction).



Step 2: ROI Extraction

- COCO bounding boxes define regions of interest - currently, parking-space regions for PARKS project integration.
- Only the ROI regions are compressed and transmitted as JPEG, reducing transmission size when compared to the full frame.





JPEG Compression and LoRa Transmission

Step 3: JPEG Compression

- Compresses grayscale ROI data using the ESP32 JPEG encoder.
- Images quality balances compression and image clarity.
- Further reduces payload from ~13,000 bytes (full-frame JPEG) to ~6,600 bytes (ROI-only JPEG), achieving ~50% additional reduction before LoRa transmission.

Step 4: Packetized Transmission

- Custom message types: DAT, ACK, CMD, INF, ERR.
- Images are split into sequenced LoRa packets.
- 8-byte header + payload per packet.
- Max module payload: 240 bytes.
- Selected payload: 216 bytes for reliability.



Gateway: Reception and Reconstruction

Step 5: Receiver-Side Reconstruction

- Reassembles sequenced LoRa packets on Raspberry Pi 5.
- Decodes JPEG data into usable image regions.
- Sends reconstructed images to the PARKS detection pipeline.



ROI Configuration & Integration

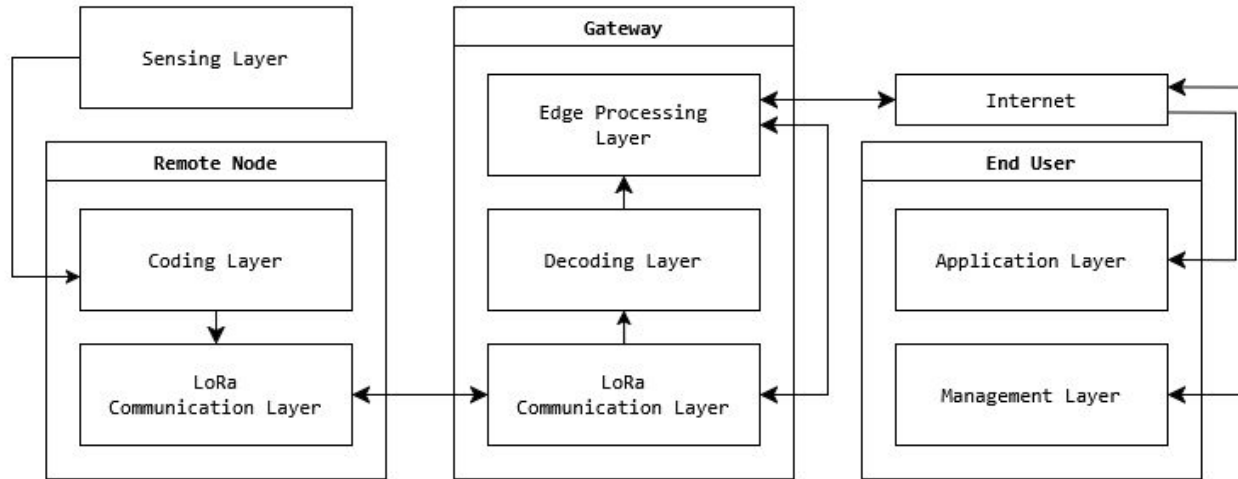
- Pre-trained YOLO model (fine-tuned on PKLot dataset) generates ROI coordinates for Remote Nodes if Gateway receives full image frame.
- Web API integrates Gateway output with ML pipelines.
- CLI + Web API tools support configuration, retrieval, and management.



Results

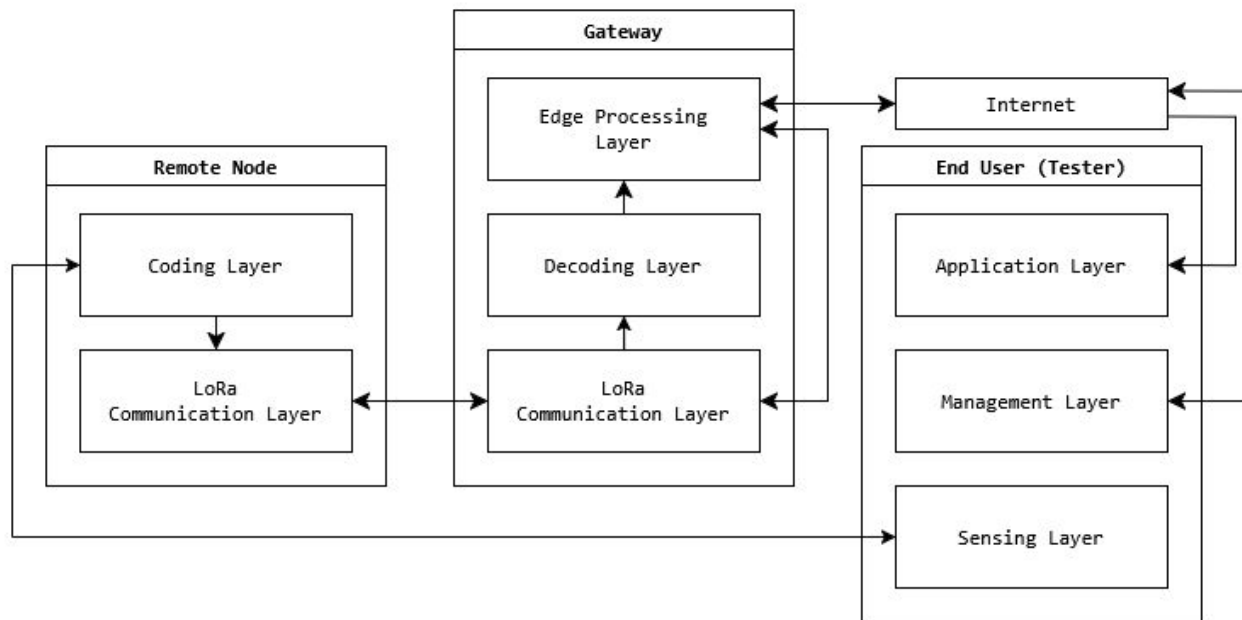


Testing Setup



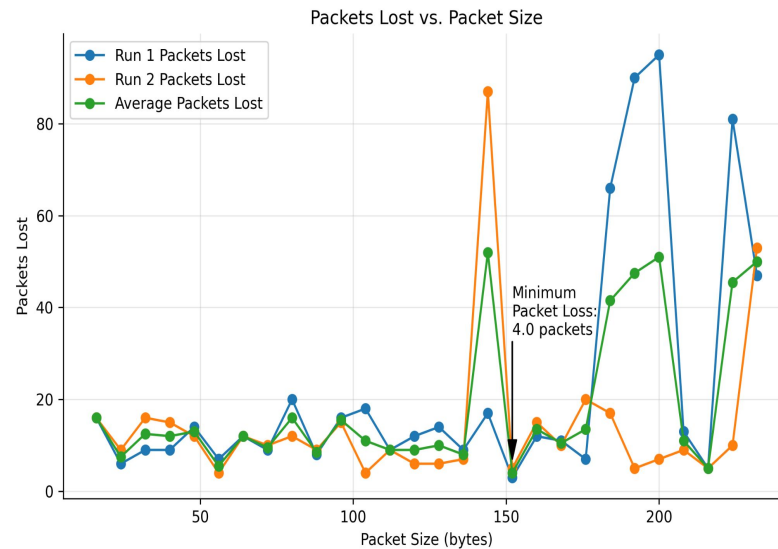
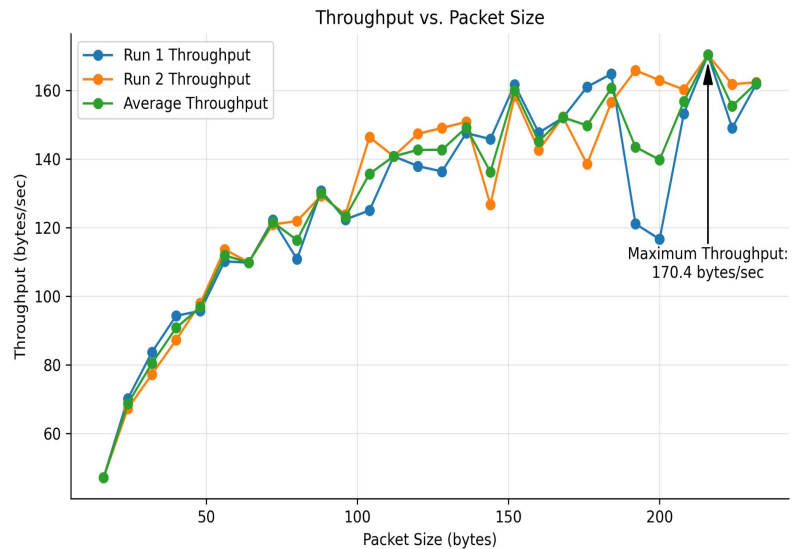


Testing Setup





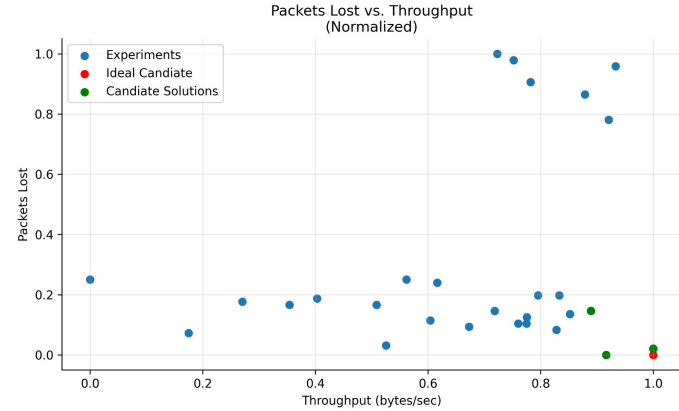
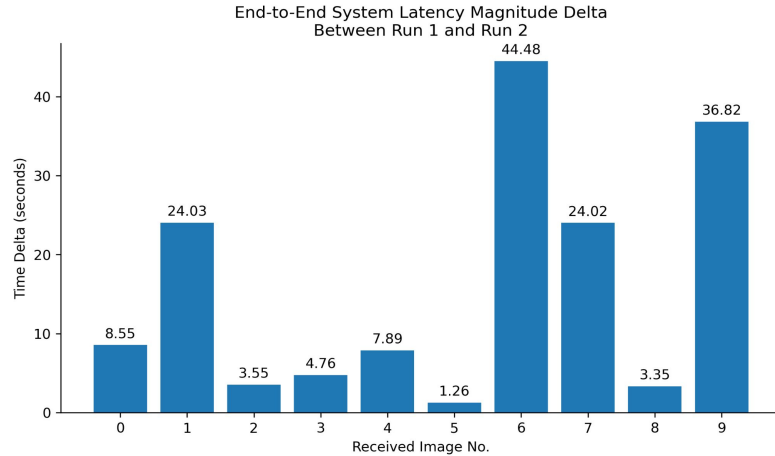
Transmission Throughput





Ideal Packet Size

- Maximize throughput.
- Minimize packet loss (retransmissions take longer than packet bursts).

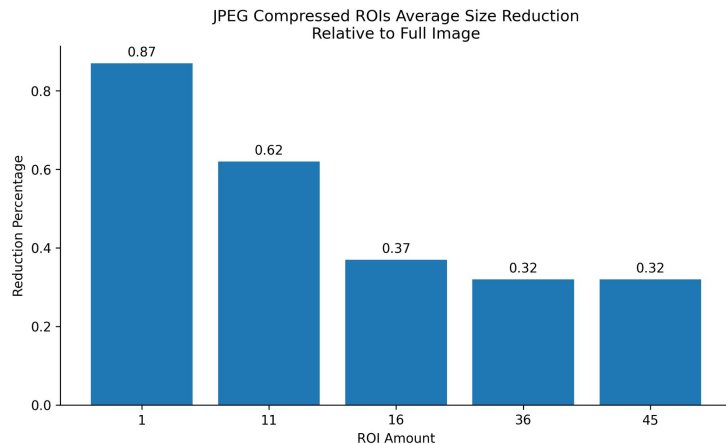
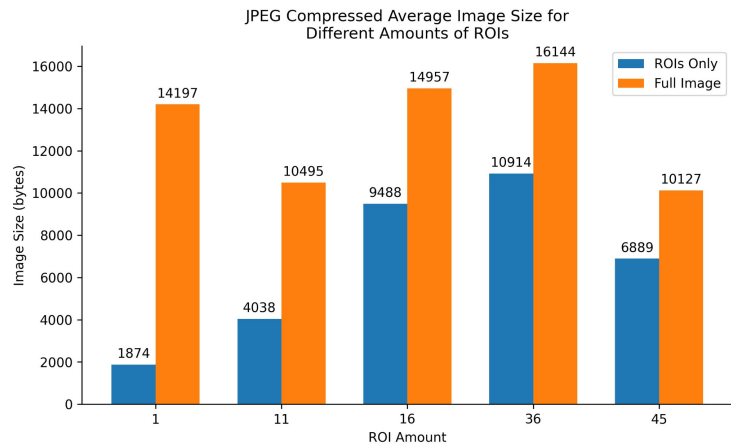
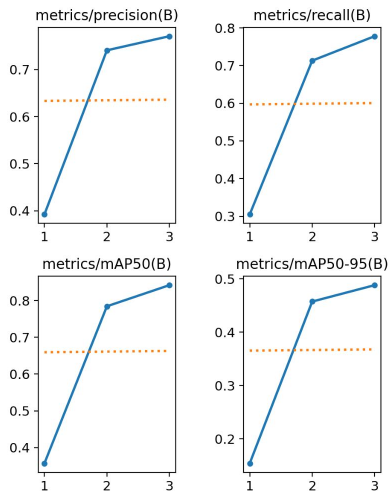


Received Image No.	Run 1 Retransmissions	Run 2 Retransmissions	End-to-end Time Delta
0	7	9	8.55
1	4	0	24.03
2	9	6	3.55
3	5	1	4.76
4	7	9	7.89
5	3	10	1.26
6	10	3	44.48
7	4	1	24.02
8	10	3	3.35
9	1	0	36.82



ROI Extraction and Masking

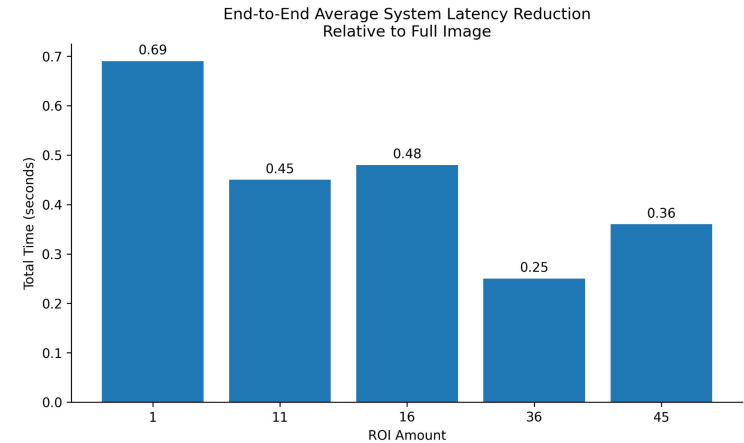
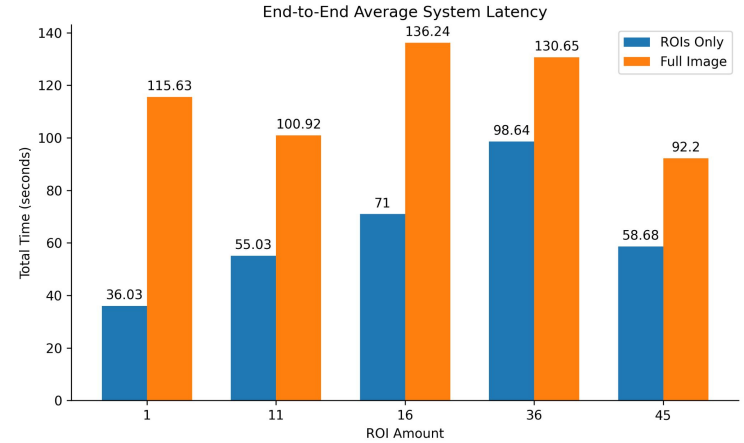
- Fine-tuned computer vision model extracts ROIs with 75% precision and recall.
- ROI masks reduced JPEG compressed QVGA image size by at least 32%.





End-to-End Performance

- Measures the time from when the Gateway receives the acknowledgement for incoming data until this data is available at the LoRaVision API.
- Best performing image end-to-end time measured under a minute, while most completely finish in under 2 minutes.





Conclusions

- Using LoRa for image data transmission is doable, especially if paired with clever approaches to reduce non-critical image data before transmission.
- Transferring full image frames of low quality images can be accomplished in around 2 minutes average, and transferring ROIs only in around 1 minute average.
- Further areas of optimization should be addressed in future implementations, regarding optimizing the LoRa radio parameters for better throughput, and identifying a better suited training dataset for the ROI extraction model.
- Minor changes in the testing approach to measure the amount of retransmission bursts instead of measuring the amount of retransmitted packets. Doing so will provide a clearer picture on the impact of retransmissions in end-to-end time.



Demo



Questions?